# An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

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## Abstract

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In vision, attention is either applied in conjunction with convolutional networks, or used to replace certain components of convolutional networks while keeping their overall structure in place. We show that this reliance on CNNs is not necessary and a pure transformer applied directly to sequences of image patches can perform very well on image classification tasks. When pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks, Vision Transformer (ViT) attains excellent results compared to state-of-the-art convolutional networks while requiring substantially fewer computational resources to train. Our largest model, ViT-H/14, has 632 million parameters, uses 16x16 patches, and was pre-trained on 32 TPUv3 cores; it reaches 88.55 percent top-1 accuracy on ImageNet when fine-tuned.

*Keywords: vision transformer, image classification, self-attention, transfer learning, patches*

## 1 Introduction

Self-attention-based architectures, in particular Transformers, have become the model of choice in natural language processing. The dominant approach is to pre-train on a large text corpus and then fine-tune on a smaller task-specific dataset. In computer vision, however, convolutional architectures remain dominant. Attention mechanisms have been added to convolutional networks, or used to replace some components, but the overall structure of the convolutional network has been kept.

In this paper we show that a pure Transformer applied directly to sequences of image patches can perform very well on image classification. We split an image into fixed-size patches, linearly embed each patch, add position embeddings, and feed the resulting sequence to a standard Transformer encoder. We call this model the Vision Transformer, or ViT. When pre-trained on large datasets and transferred to smaller benchmarks, ViT matches or outperforms state-of-the-art convolutional networks while requiring substantially fewer computational resources to train.

## 2 Related Work

Transformers were introduced for machine translation and have since become the standard architecture for language understanding. The key ingredient is self-attention, which allows each position to attend to all other positions in a sequence. Scaling Transformers to images requires a way to convert the two-dimensional image into a one-dimensional sequence, and the most natural choice is to split the image into patches.

Prior work has applied attention to images in several ways. Some models combine convolutional feature maps with attention, while others use attention to replace pooling or to model long-range dependencies. Our model differs in that it uses no convolutions at all: the patch embeddings are linear projections, and all spatial reasoning is performed by the Transformer. This makes the architecture simple and allows it to be trained with the same recipe as a language model.

## 3 Method

We follow the standard Transformer encoder as closely as possible. The model receives a sequence of patch embeddings, processes it with alternating multi-head self-attention and MLP blocks, and produces a classification prediction from a special token. The only image-specific components are the patch extraction and the position

embeddings.

### 3.1 Vision Transformer

An image is split into a grid of non-overlapping patches. For a 224x224 image and 16x16 patches, this produces 196 patches. Each patch is flattened and linearly projected to a vector of dimension d_model, and the resulting sequence is prepended with a learnable classification token. Position embeddings are added to the patch embeddings to retain spatial information, and the sequence is passed through a standard Transformer encoder with alternating multi-head attention and MLP blocks. Equation (1) shows the attention operation, and Equation (2) shows the patch embedding.

The Transformer encoder consists of L layers, each with multi-head self-attention and a position-wise MLP. Layer normalization is applied before each block, and residual connections are used around each block. The classification token output is passed through a linear layer to produce the class logits. Table 1 summarizes the three model variants, ViT-Base, ViT-Large, and ViT-Huge.

$$\text{Attention}(Q, K, V) = \text{softmax}(Q K^T / \sqrt{d_k}) V \tag{1}$$

$$z_0 = [x_{\text{class}}; E x_1; \ldots; E x_N] + E_{\text{pos}} \tag{2}$$

Table 1: ViT model variants. All models use 16x16 patches unless noted.

| Model | Layers | d_model | Heads | Params |
|---|---|---|---|---|
| ViT-Base | 12 | 768 | 12 | 86M |
| ViT-Large | 24 | 1024 | 16 | 307M |
| ViT-Huge | 32 | 1280 | 16 | 632M |

### 3.2 Fine-tuning and Few-shot Learning

We pre-train the model on a large dataset and then fine-tune it on the target dataset. For fine-tuning we replace the classification head with a new linear layer and train all parameters for a small number of epochs. We also evaluate the model in a few-shot setting, in which only a few training examples per class are available, and in a linear probe setting, in which the encoder is frozen and only a linear classifier is trained.

When the target dataset is much smaller than the pre-training dataset, fine-tuning the whole model can overfit. In that case we found it better to use a higher learning rate for the newly initialized head and a smaller learning rate for the pre-trained encoder. We also use RandAugment and mixup during fine-tuning, and we remove them when the target dataset is very small.

## 4 Experiments

We evaluate ViT on ImageNet, CIFAR-10, CIFAR-100, Oxford Flowers-102, and the VTAB suite. We first compare the model to state-of-the-art convolutional networks when pre-trained on large datasets, and then study the effect of pre-training data size, model scale, and patch size.

### 4.1 Setup

We pre-train on ImageNet-21k, which contains 14 million images in 21,843 classes, and on JFT-300M, which contains 303 million images in 18,000 classes. We use Adam with beta_1 = 0.9 and beta_2 = 0.999, a batch size of 4096, and a weight decay of 0.1, and we train for 300,000 steps with a cosine learning rate schedule. The largest model, ViT-H/14, has 632 million parameters and was trained on 32 TPUv3 cores. For fine-tuning we use SGD with momentum 0.9, a batch size of 512, and 10 epochs.

### 4.2 Comparison with State of the Art

When pre-trained on JFT-300M, ViT-H/14 achieves 88.55 percent top-1 accuracy on ImageNet when fine-tuned, which is comparable to the best convolutional networks while requiring substantially less pre-training compute. On CIFAR-100 it achieves 93.2 percent, on Oxford Flowers-102 it achieves 99.4 percent, and on the VTAB suite it achieves 77.9 percent. Table 2 summarizes the ImageNet results for the three ViT variants.

The results are particularly strong when the pre-training dataset is large. With ImageNet-21k pre-training, ViT-L/16 achieves 85.3 percent top-1 accuracy, which is slightly below the best convolutional networks, but with JFT-300M pre-training the same model reaches 87.8 percent. This suggests that the inductive bias of convolutions is useful

when data is limited, but that Transformers can learn appropriate inductive biases from sufficiently large datasets.

Table 2: ImageNet top-1 accuracy of ViT variants after fine-tuning on JFT-300M pre-training.

| Model | Patch | Top-1 |
|---|---|---|
| ViT-B/16 | 16x16 | 84.1 |
| ViT-L/16 | 16x16 | 87.8 |
| ViT-H/14 | 14x14 | 88.55 |
| ResNet-152x4 | conv | 87.5 |

### 4.3 Pre-training Data

We study how the pre-training dataset size affects performance. ViT-L/16 pre-trained on ImageNet-21k performs worse than the same model pre-trained on JFT-300M, and the gap is larger for smaller datasets. On the smaller datasets, the model benefits from stronger regularization and from a larger patch size, which reduces the sequence length and the number of parameters.

### 4.4 Scaling Study

We train models of different sizes and patch sizes on JFT-300M and measure their transfer performance. Performance improves with model size and with smaller patch size, but the computational cost grows quickly. The 16x16 patch size gives the best trade-off between accuracy and compute for the model sizes we consider, and the 32x32 patch size is cheaper but less accurate.

### 4.5 Inspecting Vision Transformer

We visualize the learned position embeddings and the attention maps. The position embeddings learn to represent the two-dimensional structure of the image, and the attention maps show that the model attends to semantically relevant regions. In the lower layers, attention often attends to nearby patches, while in the higher layers it attends to objects and their parts, which is consistent with the behavior of convolutional networks.

## 5 Conclusion

We have shown that a pure Transformer can be applied directly to images and can match or outperform state-of-the-art convolutional networks on image classification when pre-trained on large datasets. The Vision Transformer is simple and scalable, and it opens the door to using the same architecture for vision and language. Future work should study self-supervised pre-training, video, and detection, where the benefits of large-scale pre-training may be even greater.

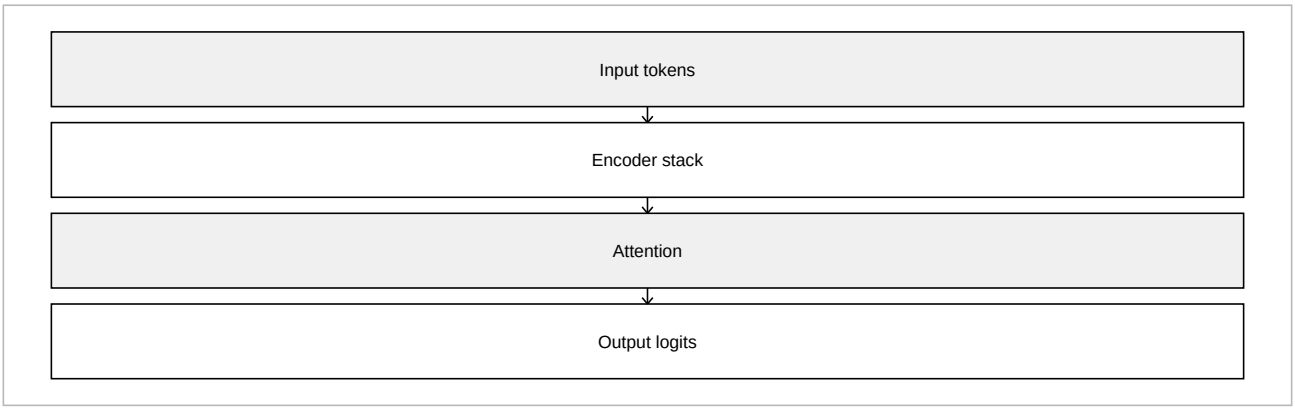


Figure 1: The Vision Transformer architecture: image patches are linearly embedded and processed by a Transformer encoder.

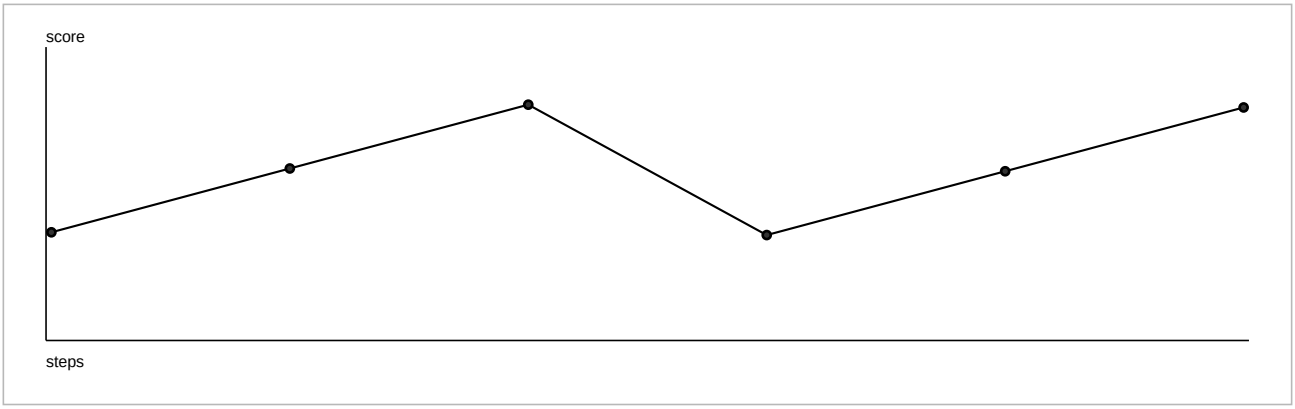


Figure 2: Attention maps from the last layer, showing attention to semantically relevant image regions.

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## A Appendix: Training Details

For pre-training we use Adam with beta_1 = 0.9, beta_2 = 0.999, a batch size of 4096, weight decay 0.1, and a cosine learning rate schedule with a linear warmup over 10,000 steps. The base learning rate is 0.003 for ViT-Base and ViT-Large, and 0.0024 for ViT-Huge. We use 16x16 patches for most experiments and 14x14 patches for ViT-H/14, which produces 256 patches for a 224x224 input.

For fine-tuning we use SGD with momentum 0.9, a batch size of 512, and 10 epochs, and we sweep the learning rate over the values 0.001, 0.003, 0.01, and 0.03. We apply RandAugment and mixup during fine-tuning, and we use a weight decay of 0.0001. The largest model, ViT-H/14, was pre-trained on JFT-300M on 32 TPUv3 cores and reaches 88.55 percent top-1 accuracy on ImageNet after fine-tuning.